

tests/ test_constitution_inheritance.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Authority:** Helix Constitution §11.4.35 (inheritance clarity), §3 (submodule inheritance propagation), §11.4 (no silent-skip PASS-bluff), §1.1 (paired mutation)

Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview / Purpose

The **comprehensive host-side inheritance test** — a superset of the gate. It asserts ALL five inheritance invariants (by delegating to `constitution_inheritance_gate.sh`, Part A) PLUS recursive nested-submodule inheritance pointers (§3, Part B). It is the file the init task's final verification points at (`bash tests/test_constitution_inheritance.sh`). Read-only.

Usage

```
bash tests/test_constitution_inheritance.sh
```

No CLI flags or arguments.

Inputs

None (no env vars, no arguments). It enumerates nested submodules via `git submodule status --recursive` and reads each owned child's `CLAUDE.md` + `AGENTS.md`. The constitution submodule itself is excluded (it is the canonical SOURCE, not a consumer — §11.4.35).

Outputs

- **Part A:** delegates to the gate; one PASS/FAIL for "all gate invariants (I1-I5) hold".
- **Part B:** for every owned nested submodule, a PASS/FAIL that its CLAUDE.md and AGENTS.md carry the Helix Constitution inheritance pointer. When there are **zero** owned children, it says so LOUDLY (a note + an explicit PASS that propagation is a verified no-op) — never a silent skip (§11.4 PASS-bluff).
- A === summary: N pass, M fail === line (fail details to stderr).
- Exit 0 — all assertions hold; exit 1 — one or more FAILED.
- Colour is emitted only when stdout is a TTY.

Side-effects

None. Read-only; runs `git submodule status --recursive` and reads files.

Dependencies

- bash (set -uo pipefail), git, grep, awk, mapfile.
- tests/constitution_inheritance_gate.sh (invoked for Part A).

Related scripts

- tests/constitution_inheritance_gate.sh — invariants I1-I5 (see docs/scripts/constitution_inheritance_gate.md).
- challenges/scripts/meta_test_constitution_inheritance.sh — the paired §1.1 mutation.
- tests/pre_build_verification.sh, tests/run-tests.sh — other callers of the underlying gate.

Last verified

2026-07-01 — documented from source; Part A (gate delegation) and Part B (recursive owned-submodule pointer check with a loud zero-children path) read directly from the script body. Not executed here.